

Control work.

1. (2 points)

Random events A, B are independent and $P(A) = 0.6$, $P(A \cup B) = 0.8$. Find

I. $P(\overline{A} \setminus B \mid A \cup \overline{B})$;

II. $P(A \setminus \overline{B} \mid \overline{A} \cup B)$.

2. (3 points)

Students draw one of 59 exam tickets. Bruce knows 8 tickets for an "excellent" grade (5), 21 for a "good" grade (4), and 24 for a "satisfactory" grade (3); the rest he does not know ("unsatisfactory" (2)). Bruce draws his ticket second.

I.

a) What is the probability that he will pass the exam with a 4 or 3?

b) Given that Bruce passed his ticket with a 4, what is the probability that he would have answered the first student's ticket with a 3?

II.

a) What is the probability that he will pass the exam with a 4 or 5?

b) Given that Bruce passed his ticket with a 3, what is the probability that he would have answered the first student's ticket with a 5?

3. (3 points)

Let's roll an asymmetric eight-sided die, numbered as follows:

I: from 4 to 11,

II: from 2 to 9,

where the probability of a face with an even number is twice as high as the probability of a face with an odd number. Let X be a random variable equal to the rolled value.

a) Construct the probability distribution of X and find $E(X)$ and $D(X)$.

b) Find the probability that the number of prime numbers:

I. lies between 240 and 300 in 1000 rolls;

II. lies between 480 and 530 in 1200 rolls.

c) Find the probability that a number divisible by 5 appears in 600 rolls:

I. exactly 120 times;

II. exactly 60 times.

4. (2 points)

Let X be a random variable with the following density:

I.
$$f(x) = \begin{cases} a(x+1), & \text{as } 0 \leq x \leq 1, \\ 0, & \text{else} \end{cases}$$

II.
$$f(x) = \begin{cases} a(2x+1), & \text{as } 0 \leq x \leq 1, \\ 0, & \text{else} \end{cases}$$

Find a , $E(X)$, the distribution function, $P(1 < X < 2.4)$ and $P(0.25 \leq X \leq 1.5)$.

5*. (1 point)

Let X and Y be random variables uniformly distributed on the interval $[a, b]$, $a > 0$. Which of the following quantities will be greater on average: the area of the rectangle formed by X and Y , or the arithmetic average of the areas of the squares formed using X and Y ?